

Grade 6 Accelerated
Unit 12
Lesson 1 Homework

Name _____
Date _____
Period _____

A map is set to a scale factor of $32 \text{ mm} = 1 \text{ km}$. Use this scale to answer questions 1-3.

1. Two mountains are 451.1 mm apart on the map. Create a proportion that could be used to determine the actual distance between the mountains.
2. Use proportions to solve for the actual distance between the mountains. Round the answer to the nearest tenth.
3. Using the scale from the map in question 1, if the distance between two towns is 25 kilometers, how many millimeters apart will the towns be on the map?

In a construction zone, there are black and gray beams that are being used for a building that are proportional in size. The black beams are 5 feet wide and 16 feet long, and the gray beams are 12.5 ft long. Use this information to answer questions 4-5.

4. Fill in the remaining values for the proportion below.

$$\frac{5}{x} = \underline{\hspace{2cm}}$$

5. Using the above proportion, determine the width of the gray beams.

Use the following information to solve problems 6-10. Suppose 8,800 people live in apartment buildings in Charlotte County, Florida. A sample of 310 people were polled asking them, if given the opportunity, would they move into a new house.

- 144 said yes.
- 150 said no.
- 16 were undecided.

6. Create a proportion to determine the total number of people who were undecided.
7. Create a proportion to determine the total number of people who said yes.
8. Create a proportion to determine the total number of people who said no.
9. To the nearest whole number, determine how many people were undecided.
10. To the nearest whole number, determine the number of people who said yes.